

Three-Dimensional Voronoi Adaptive Coverage and Dynamic Boundary Tracking for Underwater Oil Spill Monitoring

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Abstract—Underwater oil spills caused by continuous leakage form pollution plumes with three-dimensional spatial diffusion, highly nonuniform concentration distributions, and dynamically expanding boundaries. Under limited sensing ranges, conventional density-weighted centroidal coverage methods for multiple autonomous underwater vehicles may suffer from cluster collapse, where all vehicles move toward the concentration peak and lose monitoring capability over the outer plume boundary. To address this problem, this paper proposes a multi-AUV cooperative monitoring method that integrates three-dimensional Voronoi adaptive coverage with dynamic boundary tracking, referred to as CVT-DBT. First, a continuous-release three-dimensional Gaussian plume model is established by considering ocean current transport, anisotropic turbulent diffusion, buoyancy-driven rise, and natural decay. A temporal convolution integral is used to represent the cumulative contribution of historically released oil parcels. Second, the concentration field is mapped to a time-varying coverage density function, and a volume-based Voronoi partition is constructed in a bounded three-dimensional underwater domain. Furthermore, a control law consisting of centroid feedback, centroid-velocity feedforward, boundary tracking, and dispersion terms is designed based on dynamic boundary sampling and spatially balanced assignment. Simulation results show that the standard CVT method suffers from severe cluster collapse, with a final dynamic coverage ratio of 38.24% and a boundary tracking RMSE of 66.75 m. In contrast, the proposed CVT-DBT method improves the dynamic coverage ratio to 58.86% and reduces the boundary tracking RMSE to 15.94 m while satisfying the AUV velocity constraint. The results demonstrate the effectiveness of the proposed method for three-dimensional dynamic underwater oil spill monitoring.

Index Terms—Underwater oil spill monitoring, multi-AUV cooperation, three-dimensional Voronoi coverage, dynamic boundary tracking, continuous-release plume, cluster collapse.

I. INTRODUCTION

Underwater oil spills are critical threats to marine environmental safety and offshore engineering operations. Unlike surface spills, underwater leakage forms time-varying three-dimensional pollution plumes under the combined effects of ocean current transport, turbulent diffusion, buoyancy-driven rise, and natural decay. The migration and diffusion of underwater oil spills are strongly affected by the leaking rate and ocean current velocity [4], and practical oil spill modeling needs to account for transport, diffusion, decay, and boundary effects [5].

Autonomous underwater vehicles (AUVs) are suitable for underwater pollution monitoring due to their autonomous mobility, three-dimensional maneuverability, and close-range sensing capability. A key challenge in multi-AUV monitoring is to dynamically adjust the spatial distribution of vehicles according to the evolving pollution field, such that both high-concentration regions and the outer plume boundary can be monitored. Classical coverage control is typically formulated using Voronoi partitions and locational cost functions [1], while Lloyd-type iterations and centroid feedback provide numerical mechanisms for approaching centroidal Voronoi configurations [2].

However, standard centroidal Voronoi tessellation (CVT) methods are insufficient for underwater oil spill monitoring. Since the concentration peak is usually located near the leakage source, pure density-weighted centroid feedback tends to attract all AUVs toward the high-concentration core. Although this behavior may reduce the locational cost, it sacrifices the monitoring capability over the expanding outer boundary. This paper refers to this behavior as cluster collapse.

To overcome this limitation, this paper proposes a three-dimensional Voronoi adaptive coverage and dynamic boundary tracking method for multi-AUV underwater oil spill monitoring. The main contributions are summarized as follows:

- A continuous-release three-dimensional advection–diffusion–decay plume model is developed using temporal convolution of historically released oil parcels.
- A volume-based three-dimensional Voronoi coverage model is formulated, where AUVs move inside the underwater domain rather than on a two-dimensional surface.
- A dynamic boundary sampling and spatially balanced assignment mechanism is introduced, leading to a CVT-DBT control law with centroid feedback, feedforward compensation, boundary tracking, and dispersion terms.
- Comparative simulations against standard CVT and three-dimensional lawnmower coverage demonstrate the advantages of the proposed method in dynamic coverage ratio, boundary tracking RMSE, and velocity-constrained feasibility.

II. RELATED WORK

A. Multi-Agent Voronoi Coverage Control

Multi-agent coverage control aims to coordinate multiple mobile sensors or robots to achieve effective monitoring over a target domain. Cortes et al. established a fundamental coverage control framework based on Voronoi partitions and locational cost functions [1]. In this framework, each agent is responsible for the region closest to itself and moves toward the density-weighted centroid of its Voronoi cell.

Centroidal Voronoi tessellation and Lloyd algorithms provide important computational foundations for coverage control. Du et al. studied the convergence of Lloyd algorithms for computing centroidal Voronoi tessellations [2]. Nevertheless, conventional CVT methods are mainly designed for static or slowly varying density fields. When the density field is highly nonuniform, agents tend to concentrate around high-density regions, which may lead to insufficient boundary monitoring in oil spill scenarios.

B. Time-Varying Density Coverage Control

In practical environmental monitoring tasks, the density function is often time-varying. Pierson et al. studied generalized coverage control for time-varying density functions [3]. These studies indicate that pure feedback to the instantaneous centroid may be insufficient in dynamic environments, and feedforward or prediction mechanisms can improve tracking performance. In contrast to general time-varying coverage, underwater oil spill monitoring requires not only coverage of high-concentration regions but also persistent tracking of the expanding outer boundary.

C. Underwater Oil Spill Modeling

Underwater oil spill migration and diffusion are affected by leakage rate, ocean current velocity, turbulent diffusion, buoyancy, and natural decay. Previous studies have analyzed the influence of oil leaking rate and ocean current velocity on underwater spill migration [4]. Fingas provided an overview of oil spill modeling and simulation for both surface and subsurface applications [5]. Motivated by these physical insights, this paper develops a continuous-release three-dimensional plume model and integrates it with multi-AUV coverage control.

III. PROBLEM FORMULATION

Let the three-dimensional monitoring domain be

$$\Omega = [x_{\min}, x_{\max}] \times [y_{\min}, y_{\max}] \times [z_{\min}, z_{\max}] \subset \mathbb{R}^3, \quad (1)$$

where $z = 0$ denotes the sea surface and $z < 0$ denotes the underwater region. A spatial point is denoted by

$$q = [x, y, z]^T \in \Omega, \quad dq = dx dy dz. \quad (2)$$

Consider n AUVs with positions

$$p_i(t) = [x_i(t), y_i(t), z_i(t)]^T \in \Omega, \quad i = 1, \dots, n. \quad (3)$$

Each AUV satisfies the velocity constraint

$$\|\dot{p}_i(t)\| \leq v_{\max}. \quad (4)$$

Given the AUV position set $P(t) = \{p_1(t), \dots, p_n(t)\}$, the three-dimensional Voronoi cell of the i -th AUV is defined as

$$V_i(t) = \{q \in \Omega \mid \|q - p_i(t)\| \leq \|q - p_j(t)\|, \forall j \neq i\}. \quad (5)$$

Thus, $V_i(t)$ is a volume region rather than a two-dimensional plane or surface.

This partition follows directly from the locational coverage objective. For any spatial point $q \in \Omega$, the local sensing cost contributed by the i -th AUV is proportional to $\|q - p_i(t)\|^2$. Assigning q to the nearest AUV minimizes this local term pointwise, which yields the nearest-neighbor partition above in the three-dimensional Euclidean domain. Therefore, the global coverage cost can be decomposed over the three-dimensional Voronoi volume cells instead of over a planar projection or plume surface.

Let $C(q, t)$ denote the oil concentration field. The coverage density function is defined as

$$\phi(q, t) = \alpha C_{\text{norm}}(q, t) + \beta, \quad (6)$$

where

$$C_{\text{norm}}(q, t) = \min \left(\frac{C(q, t)}{C_{\text{max}}^{\text{est}}}, 1 \right). \quad (7)$$

The density-weighted load of the i -th Voronoi cell is

$$M_i(t) = \int_{V_i(t)} \phi(q, t) dq, \quad (8)$$

and the corresponding density-weighted centroid is

$$c_i(t) = \frac{\int_{V_i(t)} q \phi(q, t) dq}{\int_{V_i(t)} \phi(q, t) dq}. \quad (9)$$

The locational coverage cost is defined as

$$\mathcal{H}(P, t) = \sum_{i=1}^n \int_{V_i(t)} \phi(q, t) \|q - p_i(t)\|^2 dq. \quad (10)$$

For fixed Voronoi cells and a frozen density field, differentiating the i -th cell cost with respect to p_i gives

$$\frac{\partial}{\partial p_i} \int_{V_i(t)} \phi(q, t) \|q - p_i(t)\|^2 dq = 2M_i(t)(p_i(t) - c_i(t)). \quad (11)$$

Hence the stationary condition is $p_i(t) = c_i(t)$, which motivates the centroid-seeking update used in CVT-based coverage control.

IV. CONTINUOUS-RELEASE THREE-DIMENSIONAL OIL PLUME MODEL

Let the leakage source be located at $q_s = [x_s, y_s, z_s]^T$, where $z_s < 0$. The concentration field $C(q, t)$ satisfies the three-dimensional advection–diffusion–decay equation:

$$\begin{aligned} \frac{\partial C}{\partial t} + u \frac{\partial C}{\partial x} + w_b \frac{\partial C}{\partial z} \\ = D_x \frac{\partial^2 C}{\partial x^2} + D_y \frac{\partial^2 C}{\partial y^2} + D_z \frac{\partial^2 C}{\partial z^2} - \lambda C + Q \delta(q - q_s). \end{aligned} \quad (12)$$

A continuous leakage process is represented as the temporal convolution of historically released oil parcels. Let τ be the release time and

$$a = t - \tau \quad (13)$$

be the age of the released parcel. The center of the parcel evolves as

$$x_c(a) = x_s + ua, \quad y_c(a) = y_s, \quad z_c(a) = \min(0, z_s + w_b a). \quad (14)$$

The diffusion scales are

$$\sigma_k(a) = \sqrt{\sigma_{k0}^2 + 2D_k a}, \quad k \in \{x, y, z\}. \quad (15)$$

The nonzero initial diffusion scale $\sigma_{k0} > 0$ regularizes the point-source model and avoids concentration singularity.

The concentration field is approximated by

$$\begin{aligned} C(q, t) = & \int_0^t \frac{Qe^{-\lambda a}}{(2\pi)^{3/2} \sigma_x(a) \sigma_y(a) \sigma_z(a)} \\ & \times \exp \left[-\frac{(x - x_c(a))^2}{2\sigma_x^2(a)} \right] \\ & \times \exp \left[-\frac{(y - y_c(a))^2}{2\sigma_y^2(a)} \right] \Gamma_z(z, a) d\tau, \end{aligned} \quad (16)$$

where

$$\begin{aligned} \Gamma_z(z, a) = & \exp \left[-\frac{(z - z_c(a))^2}{2\sigma_z^2(a)} \right] \\ & + \exp \left[-\frac{(z + z_c(a))^2}{2\sigma_z^2(a)} \right]. \end{aligned} \quad (17)$$

The second term is an image term with respect to the sea surface and approximately satisfies the zero-flux boundary condition at $z = 0$. The seabed reflection is neglected by assuming that the source is sufficiently above the seabed and the main plume body does not reach the seabed within the considered monitoring window.

If AUV monitoring starts after a pre-leakage duration t_0 , the physical time used in the plume model is

$$t = t_0 + t_m, \quad (18)$$

where t_m is the monitoring time.

The dynamic plume boundary is defined as the concentration isosurface

$$\partial\mathcal{P}(t) = \{q \in \Omega \mid C(q, t) = \eta C_{\max}(t)\}. \quad (19)$$

V. THREE-DIMENSIONAL VORONOI ADAPTIVE COVERAGE AND DYNAMIC BOUNDARY TRACKING

At each time instant, the three-dimensional Voronoi partition and density-weighted centroids are computed from the current concentration field. To prevent cluster collapse, the proposed method further samples and assigns the dynamic plume boundary in three-dimensional space.

Let the sampled boundary point set be

$$\mathcal{B}(t) = \{b_1(t), \dots, b_m(t)\}, \quad b_l(t) \in \partial\mathcal{P}(t). \quad (20)$$

With the plume center $q_c(t)$ as the reference point, define

$$r_l(t) = b_l(t) - q_c(t), \quad (21)$$

and the corresponding unit direction

$$s_l(t) = \frac{r_l(t)}{\|r_l(t)\|}. \quad (22)$$

The boundary surface is approximately partitioned according to balanced three-dimensional solid-angle sectors. The boundary subset assigned to the i -th AUV satisfies

$$\mathcal{B}_i(t) \subset \mathcal{B}(t), \quad \bigcup_{i=1}^n \mathcal{B}_i(t) = \mathcal{B}(t), \quad \mathcal{B}_i(t) \cap \mathcal{B}_j(t) = \emptyset. \quad (23)$$

The boundary tracking target is

$$b_i^*(t) = \begin{cases} \frac{1}{N_i(t)} \sum_{b_l(t) \in \mathcal{B}_i(t)} b_l(t), & N_i(t) > 0, \\ p_i(t), & N_i(t) = 0, \end{cases} \quad (24)$$

where $N_i(t) = \text{card}(\mathcal{B}_i(t))$.

The nominal CVT-DBT velocity command is

$$\begin{aligned} v_i^{\text{nom}}(t) = & k_c(c_i(t) - p_i(t)) + \gamma_{\text{ff}} \dot{c}_i(t) \\ & + k_b(b_i^*(t) - p_i(t)) + v_{\text{sep},i}(t). \end{aligned} \quad (25)$$

The centroid velocity is approximated by

$$\dot{c}_i(t) \approx \frac{c_i(t) - c_i(t - \Delta t)}{\Delta t}. \quad (26)$$

The dispersion term is

$$v_{\text{sep},i}(t) = \sum_{j \neq i} \rho_{ij}(t) \frac{p_i(t) - p_j(t)}{\|p_i(t) - p_j(t)\|}. \quad (27)$$

The actual control velocity is obtained by

$$v_i(t) = \text{sat}_{v_{\max}}(v_i^{\text{nom}}(t)), \quad (28)$$

where

$$\text{sat}_{v_{\max}}(v) = \begin{cases} v, & \|v\| \leq v_{\max}, \\ v_{\max} \frac{v}{\|v\|}, & \|v\| > v_{\max}. \end{cases} \quad (29)$$

The position update is

$$p_i(t + \Delta t) = p_i(t) + v_i(t) \Delta t. \quad (30)$$

Algorithm 1 summarizes one discrete CVT-DBT iteration used in the simulations. The three-dimensional Voronoi integrals are approximated by mixed uniform and plume-biased importance samples, and each sample is assigned to its nearest AUV. This implementation is consistent with the volume Voronoi definition while avoiding explicit construction of clipped three-dimensional Voronoi polyhedra.

Let N_s be the number of centroid samples, N_b the number of boundary samples, n the number of AUVs, and L the number of temporal convolution substeps for plume evaluation. The dominant per-step cost is approximately $O(N_s n + N_s L + N_b L + n^2)$, where the four terms correspond to nearest-agent assignment, density evaluation at centroid samples, boundary concentration evaluation, and pairwise dispersion, respectively. In the considered experiments, $n = 8$, so the main computational burden comes from sampling-based density and boundary evaluation.

Algorithm 1 CVT-DBT Iteration for Dynamic Underwater Oil Plume Monitoring

Require: AUV positions $P(t)$, previous centroids C_{prev} , plume parameters, Δt , gains k_c , γ_{ff} , k_b , and v_{max}

Ensure: Updated positions $P(t + \Delta t)$, current centroids $C(t)$, and boundary targets $B^*(t)$

- 1: Draw mixed uniform and plume-biased samples in Ω .
 - 2: Evaluate $\phi(q, t)$ and the importance weights at all samples.
 - 3: Assign each sample to the nearest AUV to approximate $V_i(t)$.
 - 4: Compute density-weighted centroids $c_i(t)$ for all AUVs.
 - 5: **if** C_{prev} is available **then**
 - 6: Approximate $\dot{c}_i(t) \leftarrow (c_i(t) - c_{i,\text{prev}})/\Delta t$.
 - 7: **else**
 - 8: Set $\dot{c}_i(t) \leftarrow 0$.
 - 9: **end if**
 - 10: Sample boundary candidates near $C(q, t) = \eta C_{\text{max}}(t)$.
 - 11: Assign boundary samples to AUVs by balanced angular sectors around the plume center.
 - 12: Compute $b_i^*(t)$ as the mean assigned boundary point; use $p_i(t)$ if no point is assigned.
 - 13: Construct v_i^{nom} using centroid feedback, centroid-velocity feedforward, boundary tracking, and dispersion.
 - 14: Saturate v_i^{nom} by v_{max} to obtain $v_i(t)$.
 - 15: Update $p_i(t + \Delta t) \leftarrow p_i(t) + v_i(t)\Delta t$ and project it into Ω .
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A. Theoretical Property and Stability Discussion

For a static density field, or within a single control step where $\phi(q, t)$ is treated as frozen, the standard CVT centroid feedback admits a Lyapunov-like descent property. Consider the frozen locational cost

$$\mathcal{H}(P) = \sum_{i=1}^n \int_{V_i} \phi(q) \|q - p_i\|^2 dq. \quad (31)$$

For nondegenerate Voronoi cells with

$$M_i = \int_{V_i} \phi(q) dq > 0, \quad (32)$$

the classical CVT first variation gives

$$\frac{\partial \mathcal{H}}{\partial p_i} = 2M_i(p_i - c_i), \quad (33)$$

where the boundary terms induced by the Voronoi partition cancel between adjacent cells. Under the standard centroid feedback

$$\dot{p}_i = -k_c(p_i - c_i), \quad k_c > 0, \quad (34)$$

the derivative of $\mathcal{H}$ along the closed-loop trajectory satisfies

$$\begin{aligned} \dot{\mathcal{H}} &= \sum_{i=1}^n \left(\frac{\partial \mathcal{H}}{\partial p_i} \right)^T \dot{p}_i \\ &= -2k_c \sum_{i=1}^n M_i \|p_i - c_i\|^2 \leq 0. \end{aligned} \quad (35)$$

Therefore, in the fixed-density case, the locational cost is nonincreasing. By LaSalle's invariance principle, the standard CVT dynamics converge to the invariant set satisfying $p_i = c_i$, which corresponds to a local centroidal Voronoi configuration.

The proposed CVT-DBT law is not a pure gradient descent of $\mathcal{H}(P, t)$. The centroid feedback term inherits the local descent tendency above, whereas the centroid-velocity feedforward, boundary tracking, dispersion, and velocity saturation terms are introduced for dynamic plume monitoring and actuator feasibility. Consequently, the complete time-varying CVT-DBT system should not be claimed to be globally asymptotically convergent to the minimizer of $\mathcal{H}(P, t)$. Instead, it deliberately trades part of the locational-cost optimality for boundary tracking accuracy, which is consistent with the simulation results: standard CVT attains a lower final $\mathcal{H}$ but suffers from cluster collapse, while CVT-DBT achieves a higher dynamic coverage ratio and a much smaller boundary RMSE.

VI. SIMULATION RESULTS AND ANALYSIS

A. Simulation Setup

The simulations are conducted in MATLAB 2024b. The main parameters are listed in Table I.

TABLE I
SIMULATION PARAMETERS

Parameter	Value	Description
Number of AUVs	8	Cooperative monitoring
Source position	[150, 0, -180] m	Underwater source
Current velocity	0.50 m/s	Positive x direction
Buoyancy velocity	0.22 m/s	Positive z direction
Pre-leakage time	80 s	Initial plume exists
Simulation steps	200	1 s per step
Sensing radius	50 m	AUV sensing range
Maximum speed	3.5 m/s	Velocity constraint
Boundary threshold	3.5% C_{max}	Plume boundary

Three methods are compared: Proposed CVT-DBT, Standard CVT, and Lawnmower CPP. The standard CVT method uses only centroid feedback,

$$v_i(t) = k_c(c_i(t) - p_i(t)). \quad (36)$$

The lawnmower method follows a preplanned three-dimensional scanning path without adapting to the real-time plume state.

B. Evaluation Metrics

The coverage quality is

$$\mathcal{H}(P, t) = \int_{\Omega} \phi(q, t) \min_i \|q - p_i(t)\|^2 dq. \quad (37)$$

The dynamic coverage ratio is

$$CR(t) = \frac{\int_{\mathcal{P}(t)} \mathbf{1}(\min_i \|q - p_i(t)\| \leq R_s) dq}{\int_{\mathcal{P}(t)} dq}. \quad (38)$$

The boundary tracking RMSE is

$$RMSE(t) = \sqrt{\frac{1}{n} \sum_{i=1}^n \min_{b_l(t) \in \mathcal{B}(t)} \|p_i(t) - b_l(t)\|^2}. \quad (39)$$

The AUV speed is

$$v_i^{\text{speed}}(t) = \frac{\|p_i(t + \Delta t) - p_i(t)\|}{\Delta t}. \quad (40)$$

C. Results

Fig. 1 shows the three-dimensional plume isosurfaces and AUV trajectories. The proposed method distributes AUVs between the high-concentration core and the outer boundary, forming a spatial envelope around the plume.

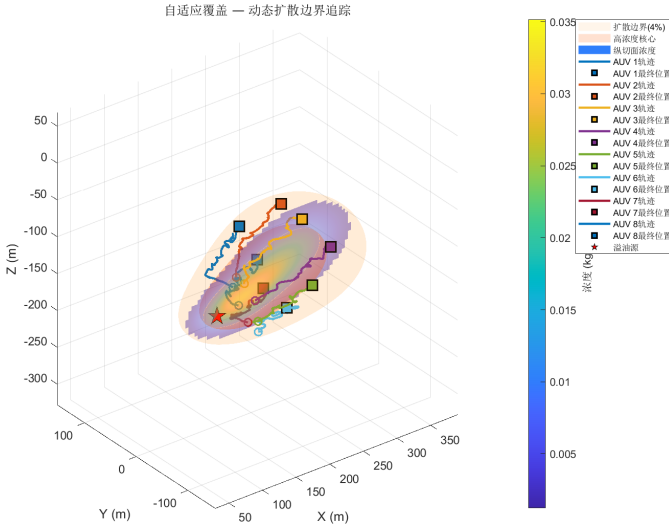


Fig. 1. Three-dimensional oil plume and AUV trajectories.

Fig. 2 shows the three-dimensional Voronoi regions, confirming that the responsibility domains are volume regions rather than two-dimensional surfaces.

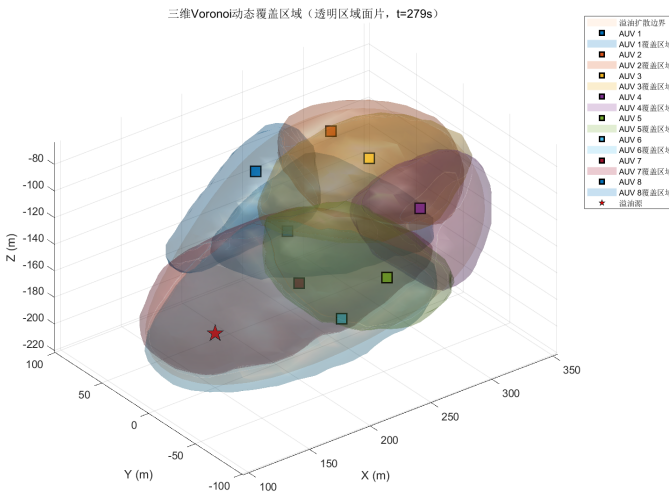


Fig. 2. Three-dimensional Voronoi coverage regions.

Fig. 3 compares the coverage quality. The standard CVT achieves a lower final coverage cost because the AUVs collapse toward the high-concentration core, but this behavior sacrifices boundary monitoring.

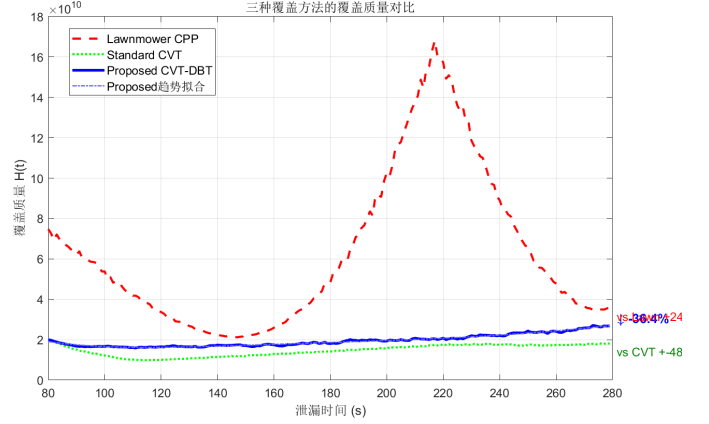


Fig. 3. Comparison of coverage quality.

Fig. 4 shows the dynamic coverage ratio and boundary tracking RMSE. The key metrics are listed in Table II.

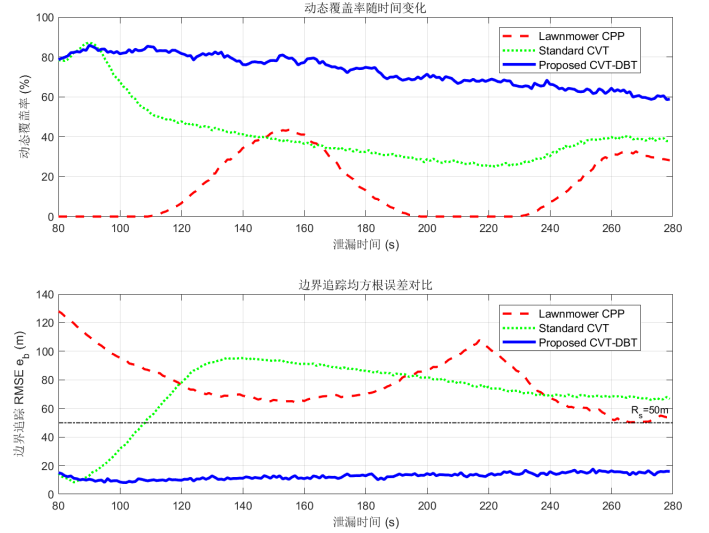


Fig. 4. Dynamic coverage ratio and boundary tracking RMSE.

TABLE II
COMPARISON OF KEY METRICS

Method	$\mathcal{H}(T)$	$CR(T)$	RMSE
CVT-DBT	2.687×10^{10}	58.86%	15.94 m
Standard CVT	1.821×10^{10}	38.24%	66.75 m
Lawnmower CPP	3.606×10^{10}	28.20%	54.10 m

Compared with standard CVT, the proposed method reduces the boundary RMSE from 66.75 m to 15.94 m, corresponding to a reduction of approximately 76.1%. Meanwhile, the dynamic coverage ratio increases from 38.24% to 58.86%. These

results indicate that dynamic boundary tracking and spatially balanced assignment effectively mitigate cluster collapse.

Fig. 5 shows the AUV speed curves. The speeds of all AUVs remain within the maximum velocity constraint of 3.5 m/s, and the curves are smooth without high-frequency oscillations.

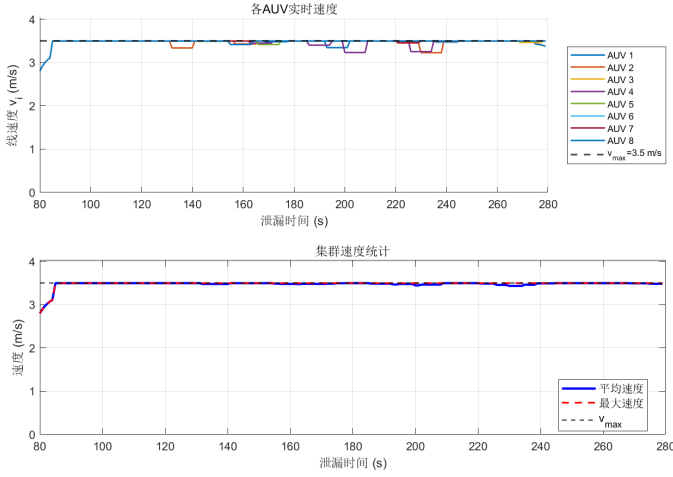


Fig. 5. AUV control input speed curves.

VII. CONCLUSION

This paper proposed a multi-AUV cooperative monitoring method that integrates three-dimensional Voronoi adaptive coverage with dynamic boundary tracking for underwater continuous oil spill monitoring. A continuous-release three-dimensional plume model was developed by considering ocean current transport, anisotropic diffusion, buoyancy-driven rise, and natural decay. A volume-based Voronoi partition and density-weighted centroid were then constructed to guide AUVs toward high-risk regions. To address cluster collapse, dynamic boundary sampling and spatially balanced assignment were introduced, leading to a velocity-constrained control law with centroid feedback, centroid-velocity feedforward, boundary tracking, and dispersion terms.

Simulation results showed that standard CVT achieved a lower locational cost but suffered from cluster collapse, resulting in a final dynamic coverage ratio of 38.24% and a boundary RMSE of 66.75 m. In contrast, the proposed CVT-DBT method improved the dynamic coverage ratio to 58.86% and reduced the boundary RMSE to 15.94 m while satisfying the AUV velocity constraint. Future work will consider collision avoidance, communication constraints, complex seabed geometry, and control barrier function based safety filtering.

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